

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026-2027

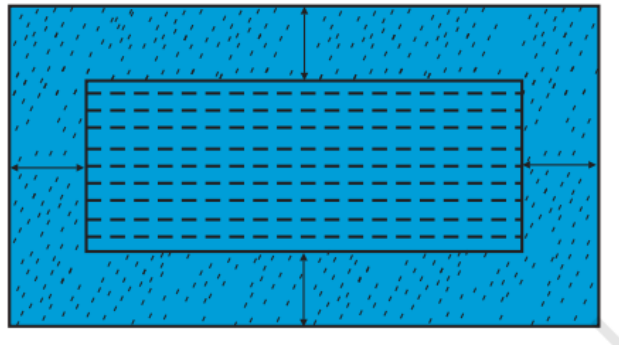
Tutorial Sheet-7

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

- (1) If $\frac{1}{2}$ is a root of the equation $x^2 + kx - \frac{5}{4} = 0$, then the value of k is
(a) 2 (b) -2
(c) $\frac{1}{4}$ (d) $\frac{1}{2}$
- (2) Which of the following equations has the sum of its roots as 3?
(a) $2x^2 - 3x + 6 = 0$ (b) $-x^2 + 3x - 3 = 0$
(c) $\sqrt{2}x^2 - \frac{3}{\sqrt{2}}x + 1 = 0$ (d) $3x^2 - 3x + 3 = 0$
- (3) Values of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is
(a) 0 only (b) 4
(c) 8 only (d) 0, 8
- (4) Which of the following equations has two distinct real roots?
(a) $2x^2 - 3\sqrt{2}x + \frac{9}{4} = 0$ (b) $x^2 + x - 5 = 0$
(c) $x^2 + 3x + 2\sqrt{2} = 0$ (d) $5x^2 - 3x + 1 = 0$
- (5) Which of the following equations has no real roots?
(a) $x^2 - 4x + 3\sqrt{2} = 0$ (b) $x^2 + 4x - 3\sqrt{2} = 0$
(c) $x^2 - 4x - 3\sqrt{2} = 0$ (d) $3x^2 + 4\sqrt{3}x + 4 = 0$
- (6) $(x^2 + 1)^2 - x^2 = 0$ has
(a) four distinct real roots (b) two distinct real roots
(c) no real roots (d) one real root.
- (7) Find the roots of the quadratic equations by using the quadratic formula in each of the following:
(i) $2x^2 - 3x - 5 = 0$ (ii) $5x^2 + 13x + 8 = 0$
(iii) $-3x^2 + 5x + 12 = 0$ (iv) $-x^2 + 7x - 10 = 0$
(v) $x^2 + 2\sqrt{2}x - 6 = 0$ (vi) $x^2 - 3\sqrt{5}x + 10 = 0$
(vii) $\frac{1}{2}x^2 - \sqrt{11}x + 1 = 0$
- (8) Find the roots of the following quadratic equations by the factorization method:
(i) $2x^2 + \frac{5}{3}x - 2 = 0$ (ii) $\frac{2}{5}x^2 - x - \frac{3}{5} = 0$
(iii) $3\sqrt{2}x^2 - 5x - \sqrt{2} = 0$ (iv) $3x^2 + 5\sqrt{5}x - 10 = 0$
(v) $21x^2 - 2x + \frac{1}{21} = 0$
- (9) Given that $\sqrt{2}$ is a zero of the cubic polynomial $6x^3 + \sqrt{2}x^2 - 10x - 4\sqrt{2}$, find its other two zeroes.
- (10) For each of the following, find a quadratic polynomial whose sum and product respectively of the zeroes are as given. Also find the zeroes of these polynomials by factorization.
(i) $-\frac{8}{3}, \frac{4}{3}$ (ii) $\frac{21}{8}, \frac{5}{16}$
(iii) $-2\sqrt{3}, -9$ (iv) $-\frac{3}{2\sqrt{5}}, -\frac{1}{2}$

- (11) The area of a rectangle is 96 cm^2 . Its length is 4 cm more than its breadth. Form a quadratic equation and find the dimensions of the rectangle.
- (12) Find a natural number whose square diminished by 84 is equal to thrice of 8 more than the given number.
- (13) If Zeba were younger by 5 years than what she really is, then the square of her age (in years) would have been 11 more than five times her actual age. What is her age now?
- (14) In the centre of a rectangular lawn of dimensions $50 \text{ m} \times 40 \text{ m}$, a rectangular pond has to be constructed so that the area of the grass surrounding the pond would be 1184 m^2 [see the figure below]. Find the length and breadth of the pond.



- (15) A train, traveling at a uniform speed for 360 km, would have taken 48 minutes less to travel the same distance if its speed were 5 km/h more. Find the original speed of the train.
- (16) A ball is thrown vertically upward with a velocity of 20 m/s. Its height after t seconds is given by:

$$h = 20t - 5t^2$$

- (a) After how many seconds will the ball hit the ground?
- (b) What is the maximum height reached?
